

2.3. Empirical, Molecular Formula and Limiting Reactants

VERY SHORT ANSWER QUESTIONS

1. [2060 Q.No. 2] What is a limiting reactant? Why is it essential in stoichiometric calculations? [2]

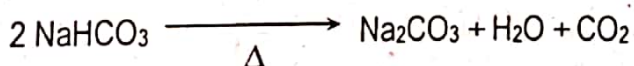
SHORT ANSWER QUESTIONS

2. [2075 Set B Q.No. 26] 2 g of magnesium is burnt in a closed vessel containing 3 g of oxygen.
- Which one is limiting reactant?
 - Calculate the mole of reactant left over.
 - How many gram of MgO are produced?
 - What is the mass of H_2SO_4 required to neutralize MgO formed in the reaction? [1+1+1+2]
- Ans: (i) Mg is limiting reagent (ii) 0.052 mol (iii) 3.33 g MgO (iv) 8.15 g H_2SO_4
3. [2071 Supp. Q.No. 24] 3.00 g of H_2 react 29.00 g of O_2 to yield H_2O . [5]
- Which is the limiting reactant?
 - Calculate the maximum amount of H_2O that can be formed.
 - Calculate the amount of reactants which remains unreacted.
- Ans: (a) H_2 is limiting reagent (b) 27 g H_2O (c) 2 g
4. [2068 Q.No. 25] A chemical reaction was carried out by mixing 25 g of pure Calcium Carbonate and 0.75 mole of pure hydrochloric acid to give CaCl_2 , H_2O and CO_2 . [1.5+1+1+1.5]
- Which one is limiting reactant and why?
 - Calculate the mass of CaCl_2 produced.
 - How many number of water molecules are formed?
 - What mass of NaOH is required to absorb the whole CO_2 produced in the reaction?
- Ans: (i) CaCO_3 (ii) 27.75 g (iii) 1.50×10^{23} (iv) 20 g
5. [2064 Q.No. 25] 10.6 g of pure Na_2CO_3 if treated with 7.9 g of HCl to produce NaCl, H_2O and CO_2 . [2+2+1]
- Find the limiting reagent and calculate mole of unreacted reagent left over.
 - What volume of CO_2 gas is produced at NTP?
 - Calculate mass of NaCl formed.

Ans: (a) Na_2CO_3 0.016 mol (b) 2.24 liter (c) 11.7 g

NUMERICAL PROBLEMS

6. [2073 Supp Q.No. 29] 2 g of magnesium is burnt in a closed vessel containing 1.2 g of oxygen to produce Magnesium oxide.
- Find the limiting reagent.
 - Calculate the number of molecules of unreacted reagent left over.
 - What mass of MgO is produced?
 - How many grams of pure HCl are required to neutralize whole MgO produced? [2+1+1+1]
- Ans: (i) O_2 (ii) 2.40×10^{21} O_2 (iii) 3 g (iv) 5.475 g
7. [2073 Set C Q. No. 29] For a reaction: [2+1+2]

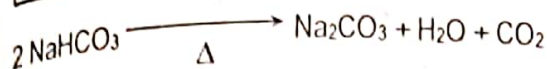


2.5 g of a sample NaHCO_3 when strongly heated gives 310 cc of CO_2 gas at 27°C and 760 mm Hg pressure.

- Calculate the percentage purity of the sample (NaHCO_3)
- How many moles of water are produced?
- What mass of pure HCl is required to neutralize Na_2CO_3 produce in the reaction?

Ans: (i) 84.4% (ii) 0.0125 mole (iii) 0.91 g

8. **2073 Set D Q. No. 29** For a reaction: [2+1+2]

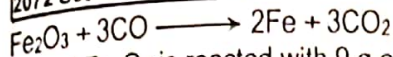


2.5 g of a sample NaHCO_3 when strongly heated gives 310 cc of CO_2 gas at 27°C and 760 mm Hg pressure.

- Calculate the percentage purity of the sample (NaHCO_3)
- How many moles of water are produced?
- What mass of pure HCl is required to neutralize Na_2CO_3 produce in the reaction?

Ans: (I) 84.4% (II) 0.0125 mole (III) 0.91 g
[1+1+1+2]

9. **2072 Set C Q.No. 26** Consider a reaction



10 g of Fe_2O_3 is reacted with 9 g of CO .

- Find the limiting reagent.
- How many mole of unreacted reactant left over?
- Calculate the mole of CO consumed in the reaction.
- What mass of NaOH is required to absorb whole CO_2 produced?

Ans: (I) Fe_2O_3 is limiting reagent (II) 0.133 mol (III) 0.1875 mol (IV) 15 g NaOH

10. **2072 Set D Q.No. 26** Consider a reaction: $\text{Fe}_2\text{O}_3 + 3\text{CO} \longrightarrow 2\text{Fe} + 3\text{CO}_2$

10 g of Fe_2O_3 is reacted with 9 g of CO .

- Find the limiting reagent.
- How many mole of unreacted reactant left over?
- Calculate the mole of CO consumed in the reaction.
- What mass of NaOH is required to absorb whole CO_2 produced?

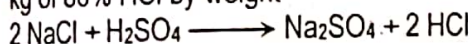
Ans: (I) Fe_2O_3 is limiting agent (II) 0.133 mol (III) 0.1875 mol (IV) 15 g NaOH
[1+1+1+2]

11. **2072 Set E Q.No. 26** 2 g of magnesium is burnt in a closed vessel containing 3 g of oxygen

- Which one is limiting reactant?
- Calculate the mole of reactant left over.
- How many gram of MgO are produced?
- What is the mass of H_2SO_4 required to neutralize MgO formed in the reaction?

Ans: (I) Mg is limiting reagent (II) 0.052 mol (III) 3.33 g MgO (IV) 8.15 g H_2SO_4
[1+1+1+2]

12. **2071 (Set C) Q.No. 24 (b)** How much sulphuric-acid containing 80% H_2SO_4 by weight is needed to produce 500 kg of 80% HCl by weight



[3]
Ans: 671.23 kg

13. **2071 (Set C) Q.No. 24 (a)** What volume of oxygen will be produced by heating 245 kg of pure KClO_3 at STP?
(atomic weight of $\text{K} = 39$)

[2]
Ans: 67200 liter

14. **2071 (Set D) Q.No. 24** 8 g of pure calcium is treated with 50 g of pure HCl to give CaCl_2 and H_2 :

- Which is limiting reactant?
- Calculate the mass of unreacted left over.
- What mass of CaCl_2 will be formed?
- How many moles of water will be produced if the whole H_2 formed in the reaction react with O_2 ? [1+1+1+2]

Ans: (a) Ca (b) 35.4 g (c) 22.2 g (d) 0.2 mol

15. **2070 Supp. Q.No. 25** 2.4 g of magnesium is treated with 0.2 mole of sulphuric acid to yield MgSO_4 and H_2 .

- Which one is limiting reactant and why?
- Calculate the mass of excess reactant.
- How many moles of MgSO_4 are produced?
- What mass of water will be produced if the whole H_2 gas formed in the reaction react with O_2 ? [1+1+1+2]

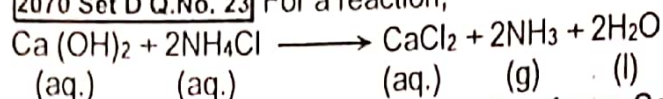
Ans: (I) Mg (II) 2.4 g (III) 0.1 mol (IV) 1.8 g

16. **2070 Set C Q.No. 28** 20 g of 40% pure CaCO_3 if reacted with 5g of HCl to produce CaCl_2 , H_2O and CO_2 .

- Find which one is limiting reactant and why?
- Calculate mass of CaCl_2 formed.
- How many number of water molecules are produced?
- Calculate the volume of CO_2 produced at 27°C and 0.5 atm pressure.

[1+1+1+2]
Ans: (a) HCl (b) 7.602g (c) 4.09×10^{22} water (d) 3.34 liter

17. **[2070 Set D Q.No. 23]** For a reaction,



The reaction is carried out by mixing 7 g of pure Ca(OH)_2 and 7 g of pure NH_4Cl .

- Find the limiting reactant.
- Calculate the mole of unreacted reactant left over.
- How many gram of CaCl_2 are formed?
- What volumes of NH_3 gas are produced at 27°C and 1.5 atmospheric pressure?

Ans: (a) NH_4Cl (b) 0.029 mol (c) 7.26g (d) 2.13 litre [1+1+1+2]

18. **[2069 (Set B) Q. No. 1]** 73 g of conc. HCl was diluted by adding 144 g of water. How many gram atom of hydrogen are present in the dilute acid?

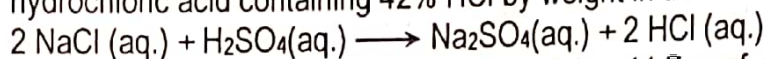
Ans: 18 gram atom [2]

19. **[2069 (Set B) Q. No. 2]** Calculate the amount of lime (CaO) that can be prepared by heating 200kg of lime stone (CaCO_3) that is 95% pure.

Ans: 106.4 kg [2]

20. 2067 Q.No. 23

- How much sulphuric acid containing 90% H_2SO_4 by weight is needed for the production of 1000 kg of hydrochloric acid containing 42% HCl by weight in the following reaction.



- If the above reaction is carried out by mixing 11.7 g of pure NaCl and 10 g of pure H_2SO_4 , find the limiting reactant.

Ans: (i) 626.483kg (ii) NaCl [3+2]

21. **[2066 Q.No. 23]** 17 g of ammonia is completely reacted with 45 g of oxygen to produce NO and H_2O . [2+1+1+1]

- Which is limiting reagent?
- Calculate the number of moles of unreacted reactant left over.
- What volume of NO are produced at NTP?
- Calculate the mass of water produced.

Ans: (i) NH_3 (ii) 0.156 mol (iii) 22.4L (iv) 27 g

22. **[2065 Q. No. 23]** 200 g of 90% pure CaCO_3 is completely reacted with excess HCl to produce CaCl_2 , H_2O and CO_2 . [1+1+1+2]

- Which one is limiting reagent?
- Calculate the mass of CaCl_2 formed.
- How many moles of water are produced?
- What volumes of CO_2 are produced if the reaction is carried out at 27°C temperature and 760mmHg pressure?

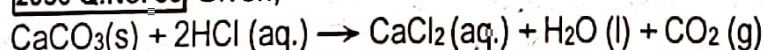
Ans: (a) CaCO_3 (b) 199.8 g (c) 1.8 (d) 44.33 litre

23. **[2062 Q.No. 23]** 5 g of pure CaCO_3 if treated with 5 g of HCl to produce CaCl_2 , H_2O and CO_2 : [5]

- Find which one is limiting reactant and why?
- Calculate mass of CaCl_2 formed.
- How many number of water molecules are produced?
- Calculate the volume of CO_2 produced at NTP.

Ans: (a) CaCO_3 (b) 5.55 g (c) 3.011×10^{22} (d) 1.12 litre [10]

24. **[2056 Q.No. 33]** Given,



If 10 grams of pure CaCO_3 are added in a solution containing 7.665 grams of HCl ,

- Find the limiting reactant.
- Calculate the number of moles excess reactant left over unreacted.
- Calculate the volume of CO_2 gas produced at NTP.
- Calculate the number of grams of NaOH required to absorb whole of the CO_2 gas as Na_2CO_3 . (atomic mass of $\text{Ca} = 40$, $\text{C} = 12$, $\text{O} = 16$, $\text{Cl} = 35.5$, $\text{Na} = 23$ and $\text{H} = 1$)

Ans: (a) CaCO_3 (b) 0.01 mol (c) 2.24 (d) 8 g